

Problem statement

You are given a small 3-class classification task and a buggy NumPy implementation of a 2-layer MLP.

Your tasks:

1. Fix the code so training works correctly.
2. Reach **validation accuracy ≥ 0.82** within **200 epochs**.
3. Print:
 - train class counts
 - validation class counts
 - validation macro-F1
4. Compare your manual gradients on one minibatch against a PyTorch model with the same weights. The **max absolute gradient difference** for each parameter should be **$< 1e-5$** .
5. Print the **top 5 validation examples with the highest loss** and explain whether they might be noisy labels.

What this tests:

- vectorized math with matrices
- softmax / cross-entropy
- backpropagation
- debugging practical ML code
- dataset analysis
- PyTorch autograd